

CSAIG10 - DATA WAREHOUSING & DATA MINING

CO2 - ASSESSMENT TOOL-2

CLASS TEST

TEJASWINI. M

192411079

BE CSE

## 1) a) DATA MINING

### DEFINITION:

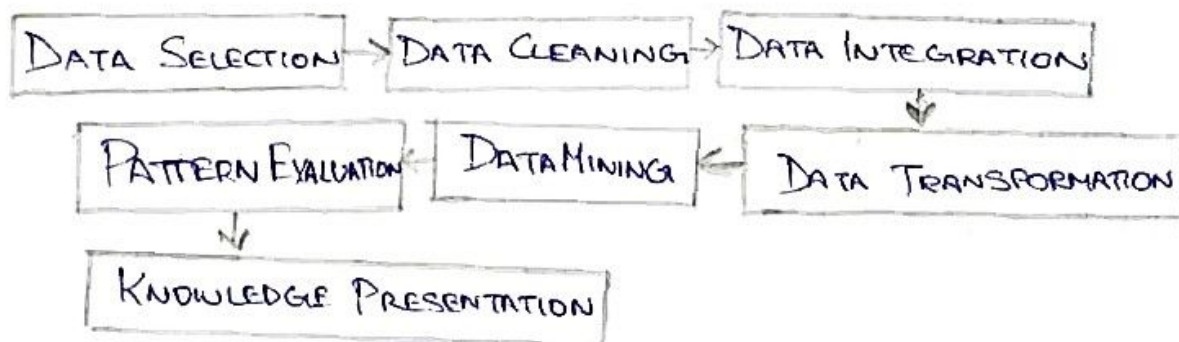
Data mining is the process of extracting useful patterns, trends, and knowledge from large volumes of data using computational techniques.

### BASIC DATA MINING TASKS

CLASSIFICATION - It assigns data into predefined categories

CLUSTERING - It groups similar data without predefined labels.

### STEPS IN THE KDD PROCESS



### ROLE OF DATA PREPROCESSING

Data preprocessing improves the quality of data by removing errors, handling missing values, and reducing noise. It increases the accuracy and efficiency of the data mining process.

## b) DATA CLEANING

### DEFINITION

Data cleaning is the process of detecting and correcting incomplete, inaccurate, or inconsistent data before analysis.

COMMON DATA QUALITY ISSUES

- MISSING VALUES
- NOISY DATA

### METHODS TO HANDLE MISSING VALUES

- Remove records if only a few values are missing.
- Replace missing values using the mean, median or mode.
- Estimate missing values using prediction techniques.

### METHODS TO HANDLE NOISY DATA

- BINNING: Groups similar values to reduce noise
- REGRESSION: Replaces noise values using a fitted model
- CLUSTERING: Detects and removes outliers.

## 2) a) DATA INTEGRATION

### DEFINITION

Data Integration is the process of combining data from multiple sources into a single, consistent dataset.

CHALLENGES

- DATA REDUNDANCY
- DATA INCONSISTENCY

### HANDLING REDUNDANCY

Duplicate records are identified and removed using

primary keys or duplicate detection methods.

## HANDLING INCONSISTENCY

Data is standardized by converting different formats and applying validation rules.

## b) COVARIANCE ANALYSIS

### DEFINITION

Covariance analysis measures how two variables change together.

- A positive covariance indicates that both variables increase or decrease together.
- A negative covariance indicates that one variable increases while the other decreases.
- A covariance of zero indicates no relationship.

### FORMULA:

POPULATION COVARIANCE 
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

SAMPLE COVARIANCE 
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

## 3) a) ARE THE STOCK PRICES INDEPENDENT?

The values show that as the price of Stock A increases, the price of Stock B also increases.

Therefore, the covariance is positive, indicating that the stock prices move together.

## 1) a) FIND COVARIANCE

GIVEN:

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean of } x = 6.2$$

$$\text{Mean of } y = 5.8$$

$$\text{Population Covariance} = -0.36$$

$$\text{Sample Covariance} = -0.45$$

## b) DESCRIBE THE MAJOR STEPS IN DATA PREPROCESSING

1) DATA CLEANING    2) DATA INTEGRATION

3) DATA TRANSFORMATION    4) DATA REDUCTION

5) DATA DISCRETIZATION

The preprocessing step improve data quality by removing errors, combining data, transforming formats, reducing data size, and converting continuous values into discrete intervals. These steps improve the accuracy and efficiency of data mining.

## 5) a) EQUAL-FREQUENCY BINNING

The given data is first arranged in ascending order and divided into bins combining an equal

## b) TYPICAL DATA MINING SYSTEM

### COMPONENTS

#### DATA SOURCES

These contain the raw data collected from different databases.

#### DATABASE OR DATA WAREHOUSE SERVER

It stores and manages the integrated data.

#### DATA CLEANING AND INTEGRATION MODULE

This module removes errors and combines data from multiple sources.

#### KNOWLEDGE BASE

It stores domain knowledge and user-defined credentials.

#### DATA MINING ENGINE

It applies algorithms such as classification, clustering and association rule mining.

#### PATTERN EVALUATION MODULE

It identifies useful and meaningful patterns.

#### GRAPHICAL USER INTERFACE (GUI)

It allows user to interact with the system and view the results.

number of values. The mean of each bin is calculated, and every value in that bin is replaced with the corresponding mean. This process smooths the data and reduces noise.

## b) DATA DISCRETIZATION

### DEFINITION

Data discretization is the process of converting continuous data into finite number of intervals or bins.

### IMPORTANCE

- It reduces data complexity
- It removes noise from the dataset
- It improves the efficiency of data mining algorithms.
- It simplifies data analysis

| EQUAL-WIDTH BINNING                     | EQUAL-FREQUENCY BINNING                                        |
|-----------------------------------------|----------------------------------------------------------------|
| Divides data into equal-sized intervals | Divides data so that each bin contains an equal no. of values. |
| Simple to implement                     | Suitable for unevenly distributed data                         |
| Sensitive to outliers                   | less affected by outliers                                      |